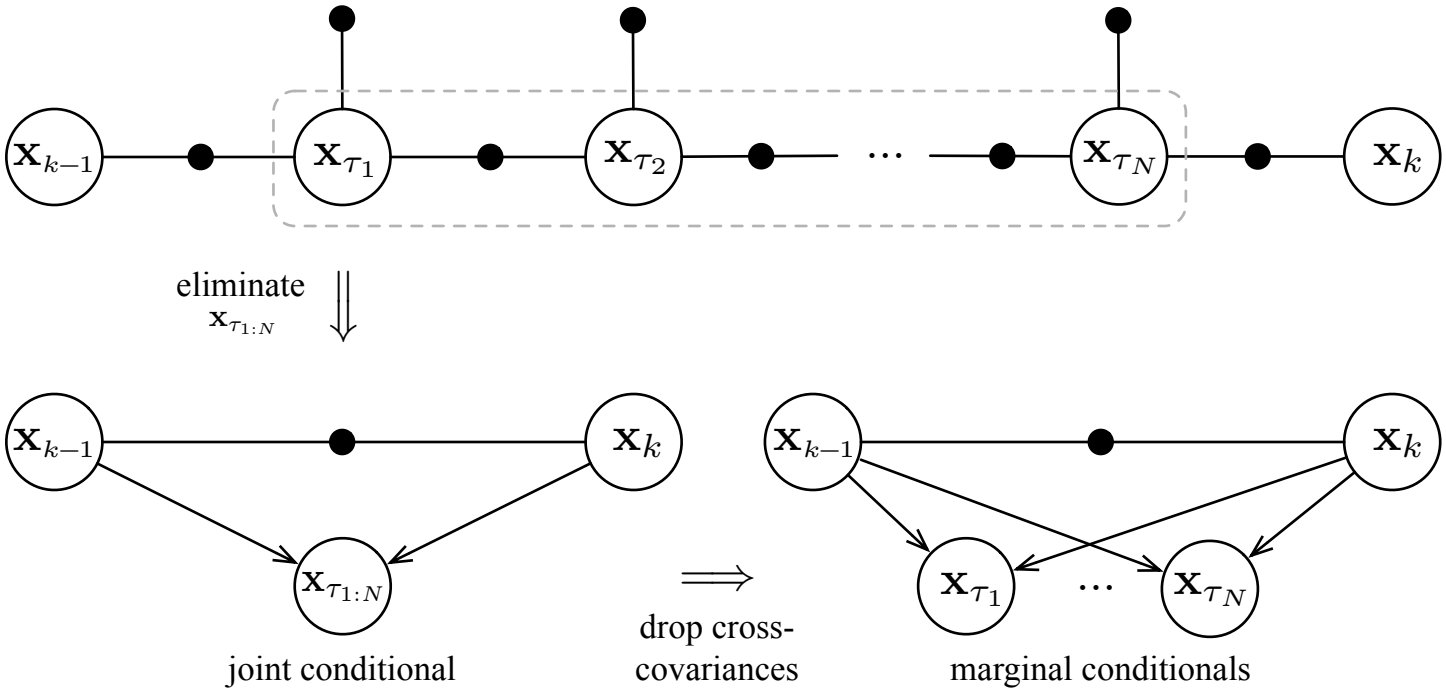
$\mathbf{x}_{k-1}$
$\mathbf{x}_{\tau_1}$
$\mathbf{x}_{\tau_2}$
$\cdots$
$\mathbf{x}_{\tau_N}$
$\mathbf{x}_k$
eliminate
$\mathbf{x}_{\tau_{1:N}}$
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\mathbf{x}_{\tau_{1:N}}$
joint conditional
drop cross-
covariances
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\mathbf{x}_{\tau_1}$
$\cdots$
$\mathbf{x}_{\tau_N}$
marginal conditionals